

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.15.15	Constant ProductionSpeed	0.0	100.0	%	80.0 ⁽¹⁾	1513	
P3.15.16	Running PumpsLimit	1	P3.15.2		3 ⁽¹⁾	1187	
M3.15.17	Interlock Signals	See Table 91 .					
M3.15.18	Overpressure Supervision	See Table 92 .					
M3.15.19	Pump Running Time	See Table 93 .					
M3.15.20	Advanced Settings	See Table 94 .					

¹ The selection of the application with parameter P1.2 Application gives the default value. See the default values in [11.25 Default Values of Parameters in the Different Applications](#).

Table 91: Interlock Signals

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.15.17.1	Pump 1 interlock	Varies	Varies		DigINSlot0.1	426	OPEN = Not active CLOSED = Active
P3.15.17.2	Pump 2 interlock	Varies	Varies		DigINSlot0.1	427	OPEN = Not active CLOSED = Active
P3.15.17.3	Pump 3 interlock	Varies	Varies		DigINSlot0.1	428	OPEN = Not active CLOSED = Active
P3.15.17.4	Pump 4 interlock	Varies	Varies		DigINSlot0.1	429	OPEN = Not active CLOSED = Active
P3.15.17.5	Pump 5 interlock	Varies	Varies		DigINSlot0.1	430	OPEN = Not active CLOSED = Active
P3.15.17.6	Pump 6 interlock	Varies	Varies		DigINSlot0.1	486	OPEN = Not active CLOSED = Active
P3.15.17.7	Pump 7 interlock	Varies	Varies		DigINSlot0.1	487	OPEN = Not active CLOSED = Active
P3.15.17.8	Pump 8 interlock	Varies	Varies		DigINSlot0.1	488	OPEN = Not active CLOSED = Active

Table 92: Overpressure Supervision Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.15.18.1	Enable Overpressure Supervision	0	1		0	1698	0 = Disabled 1 = Enabled
P3.15.18.2	Supervision Alarm Level	P3.13.1.5	P3.13.1.6	P3.13.1.4	0.00	1699	

Table 93: Pump Running Time Counter Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.15.19.1	Set Run-time Counter	0	1		0	1673	0 = No action 1 = Set the value that is specified by P3.15.19.2 to the run-time counter of the selected pump.
P3.15.19.2	Set Run-time Counter: Value	0	300 000	h	0	1087	
P3.15.19.3	Set Run-time Counter: Pump Selection	0	8		1	1088	0 = All Pumps 1 = Pump (1) 2 = Pump 2 3 = Pump 3 4 = Pump 4 5 = Pump 5 6 = Pump 6 7 = Pump 7 8 = Pump 8
P3.15.19.4	Pump RuntimeAlarm Limit	0	300 000	h	0	1109	0 = Not Used
P3.15.19.5	Pump Run-time FaultLimit	0	300 000	h	0	1110	0 = Not Used

Table 94: Advanced Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.15.22.1	Staging Frequency	P3.3.1.1	320.0	Hz	320.0	15545	
P3.15.22.2	De-staging Frequency	0.0	P3.3.1.2	Hz	0.00	15546	
P3.15.22.3	Staging Delay	0	3600	s	15588	0	0 = bandwidth delay used
P3.15.22.4	De-staging Delay	0	3600	s	15589	0	0 = bandwidth delay used

6.16 Group 3.16: Maintenance Counters

Table 95: Maintenance Counters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.16.1	Counter 1 Mode	0	2		0	1104	0 = Not used 1 = Hours 2 = Revolutions * 1000
P3.16.2	Counter 1 Alarm Limit	0	2147483647	h/kRev	0	1105	0 = Not used
P3.16.3	Counter 1 Fault Limit	0	2147483647	h/kRev	0	1106	0 = Not used
P3.16.4	Counter 1 Reset				0	1107	
P3.16.5	Counter 1 DI Reset				0	490	CLOSED = Reset

6.17 Group 3.17: Fire Mode

Table 96: Fire Mode Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.17.1	Fire Mode Password	0	9999		0	1599	1001 = Enabled 1234 = Test mode
P3.17.2	Fire Mode Frequency Source	0	18		0	1617	0 = Fire Mode frequency 1 = Preset speeds 2 = Keypad 3 = Fieldbus 4 = AI1 5 = AI2 6 = AI1 + AI2 7 = PID1 8 = Motor potentiometer 9 = Block Out.1 10 = Block Out.2 11 = Block Out.3 12 = Block Out.4 13 = Block Out.5 14 = Block Out.6 15 = Block Out.7 16 = Block Out.8 17 = Block Out.9 18 = Block Out.10
P3.17.3	Fire Mode Frequency	0.00	P3.3.1.2	Hz	50.00	1598	
P3.17.4	Fire Mode Activation on OPEN				DigIN Slot0.2	1596	OPEN = Fire Mode active CLOSED = No action
P3.17.5	Fire Mode Activation on CLOSE				DigIN Slot0.1	1619	OPEN = No action CLOSED = Fire Mode active
P3.17.6	Fire Mode Reverse				DigIN Slot0.1	1618	OPEN = Forward CLOSED = Reverse DigIN Slot0.1 = Forward DigIN Slot0.2 = Reverse
V3.17.7	Fire Mode Status	0	3			1597	0 = Disabled 1 = Enabled 2 = Activated (Enabled + DI Open) 3 = Test Mode See 5.4.2 Basic Monitoring Parameters .
V3.17.8	Fire Mode Counter	0	65535			1679	
P3.17.9	Fire Mode Run Indication Current	0	100	%	10	15580	

6.18 Group 3.18: Motor Preheat Parameters

Table 97: Motor Preheat Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.18.1	Motor Preheat Function	0	3		0	1225	0 = Not used 1 = Always in stop state 2 = Controlled by DI 3 = Temperature limit, heatsink
P3.18.2	Preheat Temperature Limit	-20	100	°C/F	0	1226	
P3.18.3	Motor Preheat Current	0	0.5*IL	A	Varies	1227	
P3.18.4	Motor Preheat ON	Varies	Varies		DigIN Slot0.1	1044	OPEN = No action CLOSED = Preheat activated in Stop state
P3.18.5	Preheat Motor Temperature	0	6		0	1045	0 = Not Used 1 = Temperature Input 1 2 = Temperature Input 2 3 = Temperature Input 3 4 = Temperature Input 4 5 = Temperature Input 5 6 = Temperature Input 6

6.19 Group 3.19: Drive Customizer

Table 98: Drive Customizer Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.19.1	Operation Mode	0	1		1	15001	0 = Execute Program 1 = Programming

6.20 Group 3.21: Pump Control

Table 99: Auto-cleaning Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.21.1.1	Cleaning Function	0	3		0	1714	0 = Disabled 1 = Enabled (DIN) 2 = Enabled (current) 3 = Enabled (weekdays)
P3.21.1.2	Cleaning Activation				DigINSlot0.1	1715	
P3.21.1.3	Cleaning Current Limit	0.0	200.0	%	120.0	1712	
P3.21.1.4	Cleaning CurrentDelay	0.0	300.0	s	60.0	1713	
P3.21.1.5	Cleaning Weekdays	0	127		0	1723	B0 = Sunday B1 = Monday B2 = Tuesday B3 = Wednesday B4 = Thursday

Index	Parameter	Min	Max	Unit	Default	ID	Description
							B5 = Friday B6 = Saturday
P3.21.1.6	Cleaning Time of Day	00:00:00	23:59:59		00:00:00	1700	
P3.21.1.7	Cleaning Cycles	1	100		5	1716	
P3.21.1.8	Clean Forward Frequency	0.00	50.00	Hz	45.00	1717	
P3.21.1.9	Clean Forward Time	0.00	320.00	s	2.00	1718	
P3.21.1.10	Clean Reverse Frequency	0.00	50.00	Hz	45.00	1719	
P3.21.1.11	Clean Reverse Time	0.00	320.00	s	0.00	1720	
P3.21.1.12	Cleaning Acceleration Time	0.1	300.0	s	0.1	1721	
P3.21.1.13	Cleaning Deceleration Time	0.1	300.0	s	0.1	1722	

Table 100: Jockey Pump Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.21.2.1	Jockey Function	0	2		0	1674	0 = Not used 1 = PID sleep 2 = PID sleep (level)
P3.21.2.2	Jockey Start Level	Varies	Varies	Varies	0.00	1675	
P3.21.2.3	Jockey Stop Level	Varies	Varies	Varies	0.00	1676	

Table 101: Priming Pump Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.21.3.1	Priming Function	0	1		0	1677	0 = Disabled 1 = Enabled
P3.21.3.2	Priming Time	0.0	320.00		3.0	1678	

Table 102: Anti-blocking Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.21.4.1	Anti-blocking Interval	0	96.0	h	0	1696	
P3.21.4.2	Anti-blocking run-time	0	300	s	20	1697	
P3.21.4.3	Anti-blocking frequency	P3.3.1.1	P3.3.1.2	Hz	15.0	1504	

Table 103: Frost Protection Parameters

Index	Parameter	Min	Max	Unit	De- fault	ID	Description
P3.21.5.1	Frost Protec- tion	0	1		0	1704	0 = Disabled

Index	Parameter	Min	Max	Unit	De- fault	ID	Description
							1 = Enabled
P3.21.5.2	Temperature Signal	0	29		6	1705	0 = Temperature Input1 (-50...200° C) 1 = Temperature Input2 (-50...200° C) 2 = Temperature Input3 (-50...200° C) 3 = Temperature Input4 (-50...200° C) 4 = Temperature Input5 (-50...200° C) 5 = Temperature Input6 (-50...200° C) 6 = Analog input 1 7 = Analog input 2 8 = Analog input 3 9 = Analog input 4 10 = Analog input 5 11 = Analog input 6 12 = ProcessDataIn1 (0–100%) 13 = ProcessDataIn2 (0–100%) 14 = ProcessDataIn3 (0–100%) 15 = ProcessDataIn4 (0–100%) 16 = ProcessDataIn5 (0–100%) 17 = ProcessDataIn6 (0–100%) 18 = ProcessDataIn7 (0–100%) 19 = ProcessDataIn8 (0–100%) 20 = Block Out.1 21 = Block Out.2 22 = Block Out.3 23 = Block Out.4 24 = Block Out.5 25 = Block Out.6 26 = Block Out.7 27 = Block Out.8 28 = Block Out.9 29 = Block Out.10
P3.21.5.3	Temperature SignalMini- mum	-50.0 (°C)	P3.21.5.4.4	°C/°F	-50.0 (°C)	1706	
P3.21.5.4	Temperature SignalMaxi- mum	P3.21.5.3	200.0 (°C)	°C/°F	200.0 (°C)	1707	
P3.21.5.5	Frost Protec- tionTempera- ture Limit	P3.21.5.3	P3.21.5.4	°C/°F	5.00 (°C)	1708	
P3.21.5.6	Frost Protec- tion Frequen- cy	0.0	P3.3.1.2	Hz	10.0	1710	
P3.21.5.7	Frost Temper- atureMonitor- ing	Varies	Varies	°C/°F		1711	This monitoring value shows the value of the temperature signal that is used for Frost Protection function.

Table 104: Flow Switch Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.21.6.1	Flow Switch				DigINSlot0.2	15590	
P3.21.6.2	Flow Switch FaultDelay	0.0	3000.0	s	20.0	15585	
P3.21.6.3	Flow Switch Bypass at Start	0.0	3000.0	s	20.0	15586	
P3.21.6.4	Flow Switch Minimum Frequency	Varies	Varies	Hz	0.0	15587	

6.21 Group 3.23: Advanced Harmonic Filter

Table 105: Advanced Harmonic Filter Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
P3.23.1	Cap Disconnect Limit	0	100	%	10	15510	
P3.23.2	Cap Disconnect Hyst	0	100	%	10	15511	
P3.23.3	AHF overtemperature				DigIN Slot0.1	15513	
P3.23.4	AHF Fault Response	0	3		3	15512	0 = No Action 1 = Alarm 2 = Fault 3 = Fault, Coast

6.22 Group 3.24: Condition-based Monitoring

Table 106: Condition-based Monitoring Parameters

Index	Name	Min	Max	Unit	Default	ID	Description
P3.24.1	Condition Based Monitoring Fault Mode	0	2		1	3540	0=No Action 1=Warnings 2=Fault + Warnings
P3.24.2 Baseline Settings							
P3.24.2.1	BaselineRun Start	0	1		0	3501	0=Disabled 1=Enabled
P3.24.2.2	BaselineRun MinFreq	Varies	Varies	Hz	0.00	3502	
P3.24.2.3	BaselineRun MaxFreq	Varies	Varies	Hz	50.00	3503	
P3.24.2.4	BaselineRun Duration	30	7200	s	120	3504	
P3.24.2.5	Modified Array	0	12		0	3505	0=Freq Points 1=Cur Max Steady 2=Cur Min Steady 3=Cur Mean Steady 4=Cur Std Steady 5=Vibr Max Ramp 6=Vibr Min Ramp 7=Vibr Mean Ramp 8=Vibr Std Ramp

Index	Name	Min	Max	Unit	Default	ID	Description
							9=Load Max Steady 10=Load Min Steady 11=Load Mean Steady 12=Load Std Steady
P3.24.2.6	Modified Point	0	9		0	3506	
P3.24.2.7	Modified Value	-300.00	320.00	various	0.00	3507	
B3.24.2.8	Modified Activation	0	1		0	3508	0=Disabled 1=Enabled
P3.24.2.9	Baseline Data Selector	0	12		0	3509	0=Freq Points 1=Cur Max Steady 2=Cur Min Steady 3=Cur Mean Steady 4=Cur Std Steady 5=Vibr Max Ramp 6=Vibr Min Ramp 7=Vibr Mean Ramp 8=Vibr Std Ramp 9=Load Max Steady 10=Load Min Steady 11=Load Mean Steady 12=Load Std Steady
P3.24.3 Stator Winding							
P3.24.3.1	Line Freq	0.00	100.00	Hz	50.00	1913	
P3.24.3.2	Line Freq Hyst	0.00	10.00	Hz	1.00	1914	
P3.24.3.3 Current Unbalance							
P3.24.3.3.1 Threshold Value							
P3.24.3.3.1.1	Mean Factor	-10.0	10.0.0		1.0	3511	
P3.24.3.3.1.2	Min Factor	-10.0	10.0.0		0.0	3512	
P3.24.3.3.1.3	Max Factor	-10.0	10.0.0		0.0	3513	
P3.24.3.3.1.4	Std Factor	-10.0	10.0.0		0.0	3514	
P3.24.3.3.1.5	Interpolation Type	0	1		1	3515	0=Constant 1=Linear
P3.24.3.3.2 Limits							
P3.24.3.3.2.1	Warning S1 Mode	0	2		0	3516	0=Abs. Value 1=Baseline Offset 2=Baseline Factor
P3.24.3.3.2.2	Warning S1 High	0.00	100.00	%	0.00	3517	
P3.24.3.3.2.3	Warning S1 Delay	0	3600	s	0	3518	

Index	Name	Min	Max	Unit	Default	ID	Description
P3.24.3.3.2.4	Warning S2 Mode	0	2		0	3519	0=Abs. Value 1=Baseline Offset 2=Baseline Factor
P3.24.3.3.2.5	Warning S2 High	0.00	100.00	%	0.00	3520	
P3.24.3.3.2.6	Warning S2 Delay	0	3600	s	0	3521	
P3.24.3.3.2.7	Alarm/Fault Mode	0	2		0	3522	0=Abs. Value 1=Baseline Offset 2=Baseline Factor
P3.24.3.3.2.8	Alarm/Fault High	0.00	100.00	%	0.00	3523	
P3.24.3.3.2.9	Alarm/Fault Delay	0	3600	s	0	3524	
P3.24.3.3.3 Counters							
P3.24.3.3.3.1	Warning S1 Counter	0.0	360.0	s	0.0	3541	
P3.24.3.3.3.2	Warning S2 Counter	0.0	360.0	s	0.0	3542	
P3.24.3.3.3.3	Alarm/Fault Counter	0.0	360.0	s	0.0	3543	
P3.24.3.3.3.4	Stop Counter Delay	0.0	320.0	s	0.0	3549	
P3.24.4 Vibration							
P3.24.4.1	Vibration Input	0	4		0	3587	0=Not Used 1=AI1 2=AI2 3=AI3 4=AI4
P3.24.4.2 Threshold Value							
P3.24.4.2.1	Mean Factor	-10.0	10.0.0		0.0	3550	
P3.24.4.2.2	Min Factor	-10.0	10.0.0		0.0	3551	
P3.24.4.2.3	Max Factor	-10.0	10.0.0		1.0	3552	
P3.24.4.2.4	Std Factor	-10.0	10.0.0		0.0	3553	
P3.24.4.2.5	Interpolation Type	0	1		0	3554	0=Constant 1=Linear
P3.24.4.3 Limits							
P3.24.4.3.1	Warning S1 Mode	0	2		0	3555	0=Abs. Value 1=Baseline Offset 2=Baseline Factor
P3.24.4.3.2	Warning S1 High	0.00	100.00	%	0.00	3556	
P3.24.4.3.3	Warning S1 Delay	0	3600	s	0	3557	
P3.24.4.3.4	Warning S2 Mode	0	2		0	3558	0=Abs. Value 1=Baseline Offset

Index	Name	Min	Max	Unit	Default	ID	Description
							2=Baseline Factor
P3.24.4.3.5	Warning S2 High	0.00	100.00	%	0.00	3559	
P3.24.4.3.6	Warning S2 Delay	0	3600	s	0	3560	
P3.24.4.3.7	Alarm/Fault Mode	0	2		0	3561	0=Abs. Value 1=Baseline Offset 2=Baseline Factor
P3.24.4.3.8	Alarm/Fault High	0.00	100.00	%	0.00	3562	
P3.24.4.3.9	Alarm/Fault Delay	0	3600	s	0	3563	
P3.24.4.4 Counters							
P3.24.4.4.1	Warning S1 Counter	0.0	360.0	s	0.0	3564	
P3.24.4.4.2	Warning S2 Counter	0.0	360.0	s	0.0	3565	
P3.24.4.4.3	Alarm/Fault Counter	0.0	360.0	s	0.0	3566	
P3.24.4.4.4	Stop Counter Delay	0.0	320.0	s	0.0	3549	
P3.24.5 Load							
P3.24.5.1 Threshold Value							
P3.24.5.1.1	Mean Factor	-10.0	10.0.0		1.0	3567	
P3.24.5.1.2	Min Factor	-10.0	10.0.0		0.0	3568	
P3.24.5.1.3	Max Factor	-10.0	10.0.0		0.0	3569	
P3.24.5.1.4	Std Factor	-10.0	10.0.0		0.0	3570	
P3.24.5.1.5	Interpolation Type	0	1		1	3571	0=Constant 1=Linear
P3.24.5.2 Limits							
P3.24.5.2.1	Warning S1 Mode	0	2		0	3572	0=Abs. Value 1=Baseline Offset 2=Baseline Factor
P3.24.5.2.2	Warning S1 Low	-300.0	300.0	%	0.00	3584	
P3.24.5.2.3	Warning S1 High	-300.0	300.0	%	0.00	3573	
P3.24.5.2.4	Warning S1 Delay	0	3600	s	0	3574	
P3.24.5.2.5	Warning S2 Mode	0	2		0	3575	0=Abs. Value 1=Baseline Offset 2=Baseline Factor
P3.24.5.2.6	Warning S2 Low	-300.0	300.0	%	0.00	3585	
P3.24.5.2.7	Warning S2 High	-300.0	300.0	%	0.00	3576	
P3.24.5.2.8	Warning S2 Delay	0	3600	s	0	3577	

Index	Name	Min	Max	Unit	Default	ID	Description
P3.24.5.2.9	Alarm/Fault Mode	0	2		0	3578	0=Abs. Value 1=Baseline Offset 2=Baseline Factor
P3.24.5.2.10	Alarm/Fault Low	-300.0	300.0	%	0.00	3586	
P3.24.5.2.11	Alarm/Fault High	-300.0	300.0	%	0.00	3579	
P3.24.5.2.10	Alarm/Fault Delay	0	3600	s	0	3580	
P3.24.5.3 Counters							
P3.24.5.3.1	Warning S1 Counter	0.0	360.0	s	0.0	3581	
P3.24.5.3.2	Warning S2 Counter	0.0	360.0	s	0.0	3582	
P3.24.5.3.3	Alarm/Fault Counter	0.0	360.0	s	0.0	3583	
P3.24.5.3.4	Stop Counter Delay	0.0	320.0	s	0.0	3549	

7 Diagnostics Menu

7.1 Faults

7.1.1 Uses of the Active Faults Submenu

When there is a fault or many faults, the display shows the name of the fault and blinks. Push [OK] to go back to the *Diagnostics* menu. The submenu *Active faults* shows the number of faults. To see the fault-time data, select a fault and push [OK].

The *Active faults* submenu can keep storage of maximum 10 faults. The submenu shows the faults in the sequence in which they occurred.

7.1.2 Reset Faults

When the drive shows a fault and stops, examine the cause of fault, and reset the fault. The fault stays active until it is reset.

There are 2 procedures to reset a fault in the local control place: with the [BACK/RESET] button and with a parameter. It is also possible to reset faults by using I/O, a fieldbus, or a PC tool.

NOTICE

Before resetting the fault, remove the external control signal to prevent restarting the drive accidentally.

7.1.2.1 Resetting Faults with the [BACK/RESET] Button

Procedure

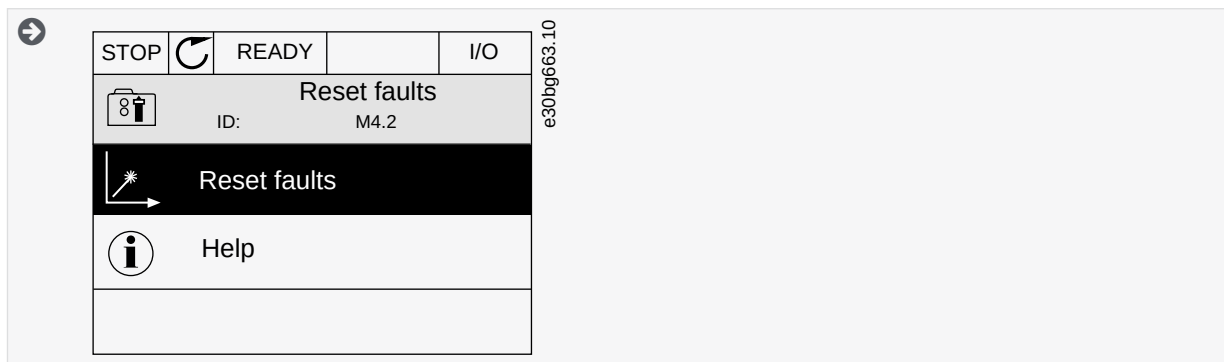
1. Push the [BACK/RESET] button on the keypad of the control panel for 2 seconds.

7.1.2.2 Resetting Faults with a Parameter on the Graphical Display

Faults can be reset with the parameter *Reset Faults*.

Procedure

1. Go to the *Diagnostics Menu*.
2. Go to the submenu *Reset faults*.
3. Select the parameter *Reset Faults*.



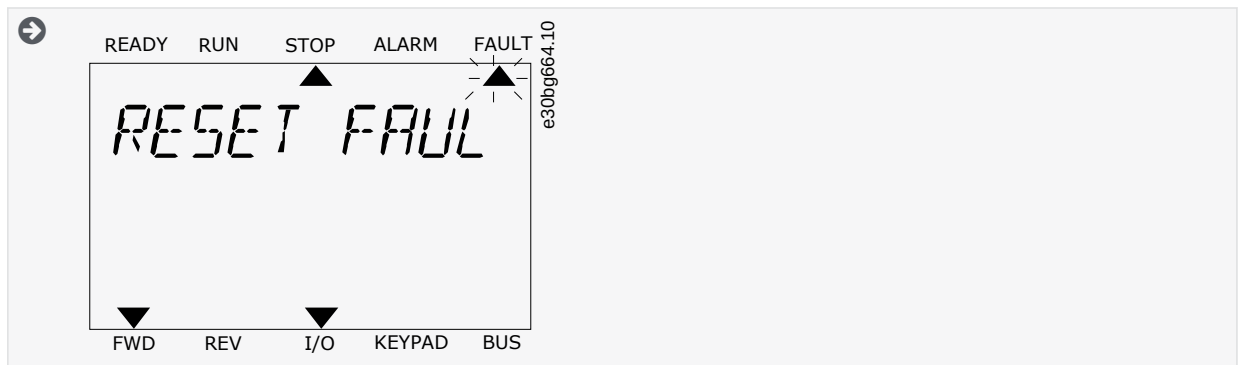
7.1.2.3 Resetting Faults with a Parameter on the Text Display

Faults can be reset with the parameter *Reset Faults*.

Procedure

1. Go to the *Diagnostics menu*.
2. Use the arrow buttons Up and Down to find the parameter *Reset Faults*.

3. Select the value Yes and push [OK].



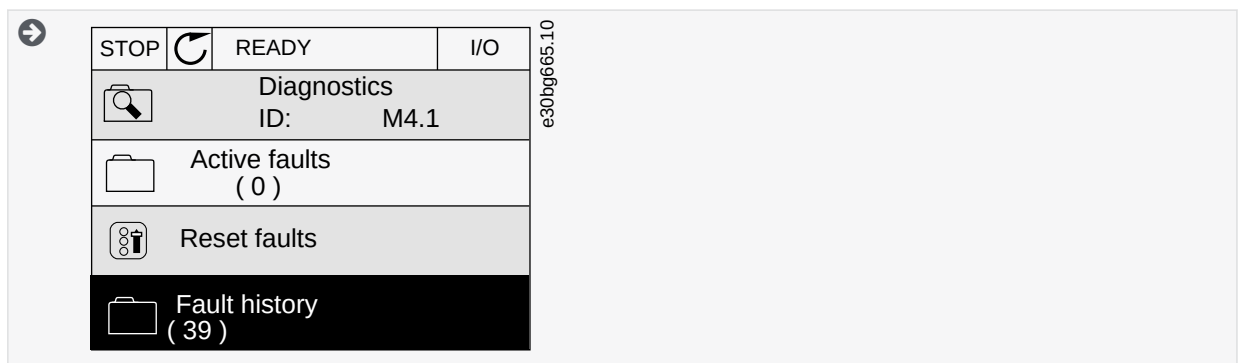
7.1.3 Uses of the Fault History

The Fault history shows more data on the faults. There is a maximum number of 40 faults in the Fault history.

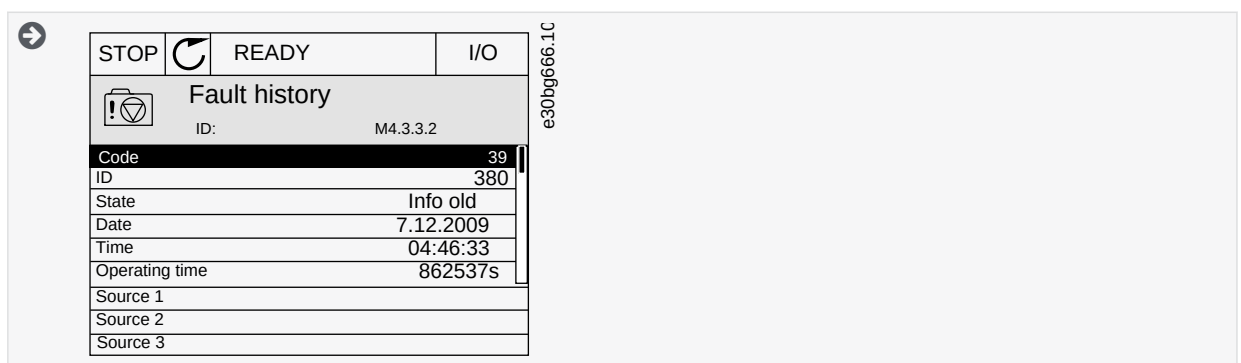
7.1.3.1 Examining the Fault History on the Graphical Display

Procedure

1. To see more data on a fault, go to *Fault history*.



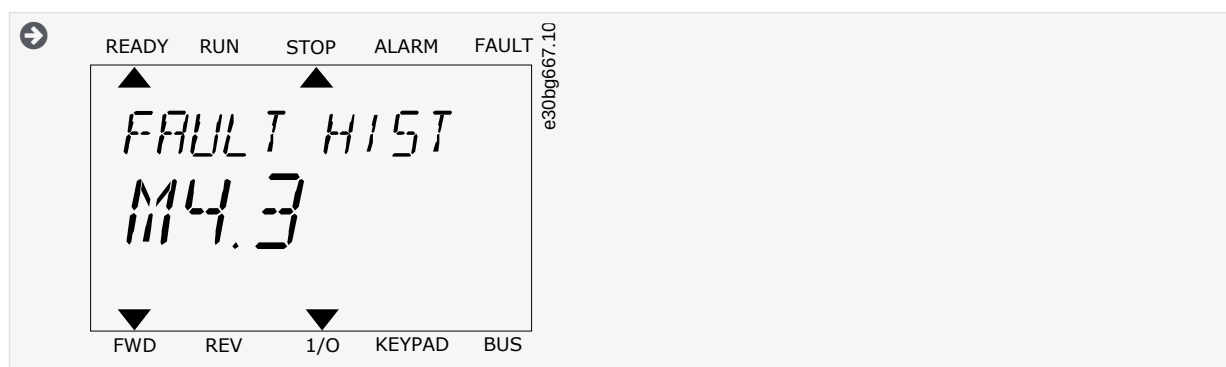
2. To examine the data of a fault, push the Arrow button Right.
3. Examine the data in a list.



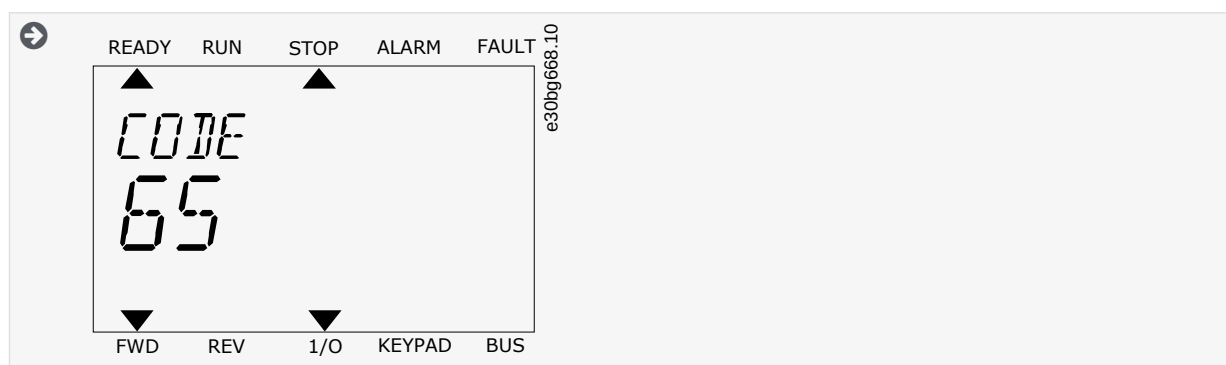
7.1.3.2 Examining the Fault History on the Text Display

Procedure

1. Push [OK] to go to *Fault history*.



2. To examine the data of a fault, push [OK] again.
3. Use the arrow button down to examine all the data.



7.2 Counters

7.2.1 Using the Counters

It is possible to monitor all the counter values from the PC, control panel, or fieldbus.

In the PC or the control panel, the counter values can be monitored in the *Diagnostics* menu. In fieldbus, see [12.4 Total and Trip Counters](#).

7.2.2 Total Counters

Table 107: The Total Counter Parameters in the *Diagnostics* Menu

Index	Parameter	Min	Max	Unit	De- fault	ID	Description
V4.4.1	Energy Counter			Varies		2291	The quantity of energy taken from the supply network. The counter cannot be reset. In the text display: the highest energy unit that the display shows is MW. If the counted energy becomes more than 999.9 MW, no unit shows on the display.
V4.4.3	Operating Time (graphical keypad)			a d hh:min		2298	The operating time of the control unit.
V4.4.4	Operating Time (text keypad)			a			The operating time of the control unit in total years.
V4.4.5	Operating Time (text keypad)			d			The operating time of the control unit in total days.
V4.4.6	Operating Time (text keypad)			hh:min:ss			The operating time of the control unit in hours, minutes, and seconds.

Index	Parameter	Min	Max	Unit	De- fault	ID	Description
V4.4.7	Run Time (graphical keypad)			a d hh:min		2293	The motor run time.
V4.4.8	Run Time (text keypad)			a			The motor run time in total years.
V4.4.9	Run Time (text keypad)			d			The motor run time in total days.
V4.4.10	Run Time (text keypad)			hh:min:ss			The motor run time in hours, minutes, and seconds.
V4.4.11	Power On Time (graphical keypad)			a d hh:min		2294	The quantity of time that the power unit has been powered on. The counter cannot be reset.
V4.4.12	Power On Time (text keypad)			a			The power on time in total years.
V4.4.13	Power On Time (text keypad)			d			The power on time in total days.
V4.4.14	Power On Time (text keypad)			hh:min:ss			The power on time in hours, minutes, and seconds.
V4.4.15	Start Command Counter					2295	The number of times that the power unit has been started.

7.2.3 Trip Counters

Table 108: The Trip Counter Parameters in the *Diagnostics Menu*

Index	Parameter	Min	Max	Unit	De- fault	ID	Description
P4.5.1	Energy Trip Counter			Varies		2296	This counter can be reset. In the text display: the highest energy unit that the display shows is MW. If the counted energy becomes more than 999.9 MW, no unit shows on the display. Resetting the counter: - In the text display: Push the [OK] button for 4 s. - In the graphical display: Push [OK]. A Reset counter page shows. Push [OK] again.
P4.5.3	Operating Time (graphical keypad)			a d hh:min		2299	This counter can be reset. See instructions in P4.5.1.
P4.5.4	Operating Time (text keypad)			a			The operating time in total years.
P4.5.5	Operating Time (text keypad)			d			The operating time in total days.
P4.5.6	Operating Time (text keypad)			hh:min:ss			The operating time in hours, minutes, and seconds.

7.3 Software Info

Table 109: The Software Info Parameters in the *Diagnostics* Menu

Index	Parameter	Min	Max	Unit	De- fault	ID	Description
V4.6.1	Software Package (graphical keypad)					2524	The code for the software identification
V4.6.2	Software Package ID (text keypad)						The code for the software identification
V4.6.3	Software Package Version (text keypad)						The code for the software identification
V4.6.4	System Load	0	100	%		2300	The load on the control unit CPU
V4.6.5	Application Name (graphical keypad)					2525	The name of the application
V4.6.6	Application ID					837	The code of the application
V4.6.7	Application Version					838	
V4.6.8	Language Selections						Selected language package in software loading
V4.6.9	Keypad Software Version						

7.4 Drive Info

Table 110: The Drive Info Parameters in the *Diagnostics* Menu

Index	Parameter	Min	Max	Unit	Default	ID	Description
V4.7.1	Power Size						

7.5 Unit Status

Table 111: The Unit Status Parameters in the *Diagnostics* Menu

Index	Parameter	Min	Max	Unit	Default	ID	Description
V4.8.1	Control Unit ID					2540	Status of the control unit
V4.8.2	Power Unit ID					2541	Status of the power unit

8 I/O and Hardware Menu

8.1 Basic I/O

Table 112: The Basic I/O Parameters in the I/O and Hardware Menu

Index	Parameter	Min	Max	Unit	Default	ID	Description
V5.1.1	Digital Input 1	0	1		0	2502	Status of the digital input signal
V5.1.2	Digital Input 2	0	1		0	2503	Status of the digital input signal
V5.1.3	Digital Input 3	0	1		0	2504	Status of the digital input signal
V5.1.4	Digital Input 4	0	1		0	2505	Status of the digital input signal
V5.1.5	Digital Input 5	0	1		0	2506	Status of the digital input signal
V5.1.6	Digital Input 6	0	1		0	2507	Status of the digital input signal
V5.1.7	Analog Input 1 Mode	1	3		3	2508	Shows the mode that is set for the analog input signal. The selection is made with a DIP switch on the control board. 1 = 0...20mA 3 = 0...10V
V5.1.8	Analog Input 1	0	100	%	0.00	2509	Status of the analog input signal
V5.1.9	Analog Input 2 Mode	1	3		3	2510	Shows the mode that is set for the analog input signal. The selection is made with a DIP switch on the control board. 1 = 0...20mA 3 = 0...10V
V5.1.10	Analog Input 2	0	100	%	0.00	2511	Status of the analog input signal
V5.1.11	Analog Output 1 Mode	1	3		1	2512	Shows the mode that is set for the analog input signal. The selection is made with a DIP switch on the control board. 1 = 0...20mA 3 = 0...10V
V5.1.12	Analog Output 1	0	100	%	0.00	2513	Status of the analog output signal
V5.1.13	Relay Output 1	0	1		0	2514	Status of the relay output signal
V5.1.14	Relay Output 2	0	1		0	2515	Status of the relay output signal
V5.1.15	Relay Output 3	0	1		0	2516	Status of the relay output signal

8.2 Option Board Slots

The parameters in this menu are different for all the option boards. The parameters of the installed option board is shown. If there is no option board in the slots C, D or E, no parameters are shown. See more about the location of the slots in chapter [11.7.1.1 General Information on the Programming](#).

When an option board is removed, the fault code 39 and the fault name *Device removed* show on the display.

Table 113: Option Board Related Parameters

Menu	Function	Description
Slot C	Settings	The settings that are related to the option board.
	Monitoring	Monitor the data that is related to the option board.
Slot D	Settings	The settings that are related to the option board.
	Monitoring	Monitor the data that is related to the option board.
Slot E	Settings	The settings that are related to the option board.
	Monitoring	Monitor the data that is related to the option board.

8.3 Real-time Clock

Table 114: The Real-time Clock Parameters in the I/O and Hardware Menu

Index	Parameter	Min	Max	Unit	De- fault	ID	Description
V5.5.1	Battery State	1	3			2205	Status of the battery. 1 = Not installed 2 = Installed 3 = Replace the battery
P5.5.2	Time			hh:mm:ss		2201	The current time of the day
P5.5.3	Date			dd.mm.		2202	The current date
P5.5.4	Year			yyyy		2203	The current year
P5.5.5	Daylight Saving	1	4			2204	The daylight saving rule 1 = Off 2 = EU: starts on the last Sunday in March, ends on the last Sunday in October 3 = US: starts on the 2nd Sunday in March, ends on the 1st Sunday in November 4 = Russia (permanent)

8.4 Simple Network Time Protocol (SNTP)

The simple network time protocol (SNTP) allows devices connected to a TCP/IP network receive precise time information from designated time servers. The time information can be received either by the client device polling the time servers or the time servers broadcasting the time.

Table 115: SNTP parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
V5.5.7.1.1	SNTP Mode					3162	
V5.5.7.1.2	Server 1 IP					3163	
V5.5.7.1.3	Server 2 IP					3164	
V5.5.7.1.4	SNTP Port					3165	
V5.5.7.1.5	Time Interval					3166	

8.5 Power Unit Settings

In this menu, the settings of the fan, the brake chopper, the sine filter, and the harmonic filter can be changed.

The fan operates in the optimized or the always on mode. In the optimized mode, the internal logic of the drive receives data about the temperature and controls the fan speed. After the drive goes in the Ready state, the fan stops in 5 minutes. In the always on mode, the fan operates in full speed, and does not stop.

The sine filter keeps the overmodulation depth in limits and does not let the thermal management functions decrease the switching frequency.

The harmonic filter can be enabled to avoid possible resonances in the DC link of the AC drive.

Table 116: Power Unit Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P5.6.1.1	Fan Control Mode	0	1		1	2377	0 = Always on 1 = Optimized
V5.6.1.5	Fan Lifetime ID					849	Lifetime of the fan of the drive
P5.6.1.6	Fan lifetime alarm limit	0	200000	h	50000	824	Alarm limit of the fan lifetime
B5.6.1.7	Fan lifetime reset ID					823	Reset the fan lifetime counter with this button.
P5.6.2.1	Brake Chopper Mode	0	3		0	2526	0 = Disabled 1 = Enabled (Run) 2 = Enabled (Run & Stop) 3 = Enabled (Run, no testing)
P5.6.3.1	Max Switching Freq	1.5	Varies	kHz		2620	
P5.6.5.1	Sine Filter	0	1		0	2527	0 = Disabled 1 = Enabled
P5.6.6.1	Harmonic Filter	0	1		0	2497	0 = Disabled 1 = Enabled

8.6 Keypad

Table 117: The Keypad Parameters in the I/O and Hardware Menu

Index	Parameter	Min	Max	Unit	De- fault	ID	Description
P5.7.1	Timeout Time	0	60	min	0	804	The time after which the display goes back to the page that is set with parameter P5.7.2. 0 = Not used
P5.7.2	Default Page	0	4		0	2318	The page that the display shows when the drive is powered up, or when the time that is set with P5.7.1 is expired. If the value is set to 0, the display shows the last page that it showed. 0 = None 1 = A menu index 2 = Main menu 3 = Control page 4 = Multimonitor

Index	Parameter	Min	Max	Unit	De- fault	ID	Description
P5.7.3	Menu Index					2499	Set a page to be the menu index. (The selection 1 in P5.7.2.)
P5.7.4	Contrast ⁽¹⁾	30	70	%	50	830	Set the contrast of the display.
P5.7.5	Backlight Time	0	60	min	5	818	Set the time after which the backlight of the display turns off. If the value is set to 0, the backlight is always on.

¹ Only available with the graphical keypad.

8.7 Fieldbus

In the I/O and Hardware menu, there are the parameters that are related to fieldbus boards. Find the instructions on how to use these parameters in the related fieldbus manual.

Table 118: Fieldbus Menu Structure

Submenu level 1	Submenu level 2	Submenu level 3	Submenu level 4
RS-485	Common settings	Protocol	Modbus RTU
			N2
			BACnet MSTP
RS-485	Modbus RTU	Parameters	Slave address
			Baud rate
			Parity type
			Stop bits
			Communication timeout
			Operate mode
		Monitoring	Fieldbus protocol status
			Communication status
			Illegal functions
			Illegal data addresses
			Illegal data values
			Slave device busy
			Memory parity error
			Slave device failure
			Last fault response
			Control Word
			Status Word
RS-485	N2	Parameters	Slave address
			Communication timeout

Submenu level 1	Submenu level 2	Submenu level 3	Submenu level 4
		Monitoring	Fieldbus protocol status
			Communication status
			Invalid data
			Invalid commands
			Command not accepted
			Control word
			Status word
RS-485	BACnet MSTP	Parameters	Baud rate
			Autobauding
			MAC address
			Instance number
			Communication timeout
		Monitoring	Fieldbus protocol status
			Communication status
			Actual instance number
			Fault code
			Control word
			Status word
Ethernet	Common settings	IP address mode	
		Duplicate IP Detection	
		Speed/Duplex	
		Ethernet Fieldbus 1	
		Ethernet Fieldbus 2	
		IP Port Filter	
		Drive PC Tool	
		Fixed IP	IP address
			Subnet mask
			Default gateway
		IP address	
		Subnet mask	
		Default gateway	
		MAC address	

Submenu level 1	Submenu level 2	Submenu level 3	Submenu level 4
Ethernet	Modbus TCP ⁽¹⁾		
Ethernet	BACnet IP ⁽²⁾		
Ethernet	EtherNet/IP ⁽³⁾		
Ethernet	PROFINET I/O ⁽⁴⁾		

¹ For details, see VACON 100 Modbus User Manual.

² For details, see VACON 100 BACnet Installation Manual.

³ For details, see VACON 100 EtherNet IP Installation Manual.

⁴ For details, see VACON 100 Profinet IO Installation Manual.

8.8 Fieldbus General

Enabling Ethernet Fieldbus Protocol

Needed internal Ethernet based fieldbus protocols can be enabled with parameters “Ethernet Fieldbus 1”, 5.9.1.4 (ID 2622) and “Ethernet Fieldbus 2”, 5.9.1.5 (ID 2623). The parameters are included since VACON® 100 INDUSTRIAL V030 and VACON® 100 FLOW V022 firmware.

Both parameters contain the same list with “PROFINET I/O, EtherNet/IP, BACnet/IP and Modbus TCP” protocols. PROFINET I/O and EtherNet/IP are visible only if included in drive configuration. By default Modbus TCP is enabled, but if not used, it is recommended to turn it off.

If the same fieldbus is selected to both parameters, the fieldbus protocol is still activated only once. Protocol specific menu tree becomes visible only after enabling it from these parameters. When using VACON® Live, to see protocol specific menus, the “5.9 Ethernet” menu tree must be reloaded after enabling the protocols.

When restoring parameters from older firmware version, drive checks on start-up which protocols were enabled in old firmware and sets up these new parameters accordingly. After having restored parameters, restart the drive. If more than two Ethernet fieldbuses were activated in old firmware, new firmware allows it. In this case, these two new parameters show “None”, but when the values are changed, all Ethernet fieldbuses are stopped, and only the enabled ones are started.

Table 119: Fieldbus Common Settings

Index	Parameter	Min	Max	Unit	Default	ID	Description
P5.9.1.4	Ethernet Fieldbus 1	None	BACnet/IP ⁽¹⁾		None	2622	2 Ethernet-based internal fieldbuses can be enabled at a time. Select fieldbus from the list.
P5.9.1.5	Ethernet Fieldbus 2	None	BACnet/IP ⁽¹⁾		Modbus TCP	2623	
P5.9.1.6	IP Port Filter	0	65535		0	2641	Use this parameter to block IP ports/protocols.
P5.9.1.7	Drive PC Tool	Configuration	Disabled		Configuration	2642	Use this parameter to block VACON® PC Tools over Ethernet, for example, VACON® Live.

¹ Depends on fieldbuses included in drive configuration

Table 120: Fieldbus General

Index	Parameter	Min	Max	Unit	Default	ID	Description
P5.13.1	Controlling Fieldbus					2539	
V5.13.2	Controlling Fieldbus					2610	

8.9 STO

Table 121: STO Parameters

Index	Parameter	Min	Max	Unit	Default	ID	Description
V5.12.1.1	Channel 1	0	1		0		See VACON® 100 OPTBJ STO Option Board User Manual.
V5.12.1.2	Channel 2	0	1		0		
P5.12.2.1	Fan Stop on STO	Disabled	Enabled		Disabled	2616	

NOTICE

These parameters are only visible when an STO option board is connected.

8.10 Fieldbus Redundancy

supports the usage of two fieldbus connections. The drive accepts control data only from the primary fieldbus. If the primary fieldbus connection fails, the control can be automatically switched to a secondary fieldbus using the fieldbus redundancy control. built-in fieldbus protocols support fieldbus redundancy control. The only exception is RS485 N2, which does not support fieldbus redundancy. Recent firmware versions of fieldbus option boards also support fieldbus redundancy.

When *switch on failing* mode is selected, and the current fieldbus connection detects missing communication from the controller, the switchover command is given to change to the other fieldbus connection. When *switch on faulted* mode is selected, and the current fieldbus connection gets a timeout, the switchover command is given to change to the other fieldbus connection.

The switchback functionality forces a change back to the primary fieldbus connection when it is in operational state and when the given time delay has elapsed. If the switchback functionality is deactivated, the switchover between the fieldbus connections is only performed when the currently controlling fieldbus begins to fail or becomes faulted.

The currently controlling fieldbus protocol can be checked from the monitoring value *V5.13.2 Controlling Fieldbus* (ID 2610).

Table 122: Fieldbus Redundancy

Index	Parameter	Min	Max	Unit	Default	ID	Description
P5.13.3.1	Redundancy Mode	0	2		0	2606	0 = Disabled 1 = Switch on failing 2 = Switch on faulted
P5.13.3.2	Primary Fieldbus Connection				0	2607	
P5.13.3.3	Secondary Fieldbus Connection				0	2608	
P5.13.3.4	Switchover Timeout	0	65535	ms	0	2609	
P5.13.3.5	Switchback Timeout	0	65535	s	10	2612	
V5.13.3.6	Switchover Counter					2611	

9 User Settings, Favourites, and User Level Menus

9.1 User Settings

9.1.1 User Settings Parameters

Table 123: General Settings in the User Settings Menu

Index	Parameter	Min	Max	Unit	Default	ID	Description
P6.1	Language selection	Varies	Varies		Varies	802	The selection is different in all the language packages.
P6.2	Application Selection					801	Select the application.
M6.5	Parameter back-up						See 9.1.2 Parameter Back-up Parameters .
M6.6	Parameter compare						See 9.1.3 Parameter Compare Parameters .
P6.7	Drive name					2528	Use VACON® Live PC tool to give a name to the drive if necessary.
M6.8	Licenses						See 9.1.4 License Parameters .

9.1.2 Parameter Back-up Parameters

Table 124: The Parameter Back-up Parameters in the User Settings menu

Index	Parameter	Min	Max	Unit	De- fault	ID	Description
B6.5.1	Restore Factory Defaults					831	Restores the default parameter values and starts the Startup wizard.
P6.5.2	Save to Keypad ⁽¹⁾	0	1		0	2478	Saves the parameter values to the control panel, for example to copy them to another drive. 0 = No 1 = Yes
P6.5.3	Restore from Key- pad ⁽¹⁾					2488	Loads the parameter values from the control panel to the drive.
B6.5.4	Save to Set 1					2489	Keeps a customized parameter set (that is, all the parameters included in the application).
B6.5.5	Restore from Set 1					2490	Loads the customized parameter set to the drive.
B6.5.6	Save to Set 2					2491	Keeps another customized parameter set (that is, all the parameters included in the application).
B6.5.7	Restore from Set 2					2492	Loads the customized parameter set 2 to the drive.

¹ Only available with the graphical display.

9.1.3 Parameter Compare Parameters

Table 125: Parameter Compare Parameters in the User Settings Menu

Index	Parameter	Min	Max	Unit	Default	ID	Description
B6.6.1	Active set - Set 1					2493	
B6.6.2	Active set - Set 2					2494	
B6.6.3	Active set - Defaults					2495	
B6.6.4	Active set - Keypad Set					2496	

9.1.4 License Parameters

Table 126: License Parameters in the User Settings Menu

Index	Parameter	Min	Max	Unit	Default	ID	Description
P6.8.1	Insert License						Insert a drive-specific license to enable new features or remove added features.

9.2 Favourites

9.2.1 Using the Favourites Menu

This menu is available on the control panel with the graphical display, but not on the control panel with the text display.

This menu is not available in the VACON® Live tool.

If the same items are used frequently, they can be added into *Favourites*. Collect a set of parameters or monitoring signals from all the control panel menus. It is not necessary to find them in the menu structure one by one. As an alternative, add them into the *Favourites* folder where it is easy to find them.

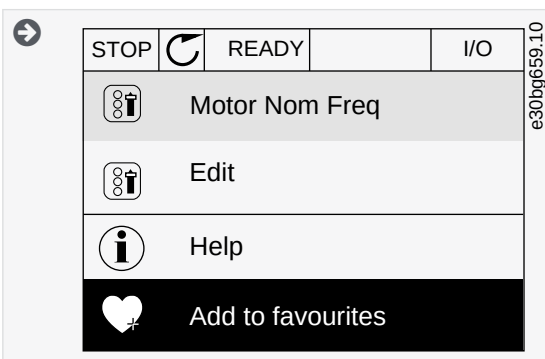
9.2.2 Adding an Item to the Favourites

It is possible to add frequently used items into *Favourites* where they are easily accessible.

Procedure

1. Find the item to be added to *Favourites*. Push the [OK] button.
2. Select *Add to favourites* and push the [OK] button.

The steps are now completed.



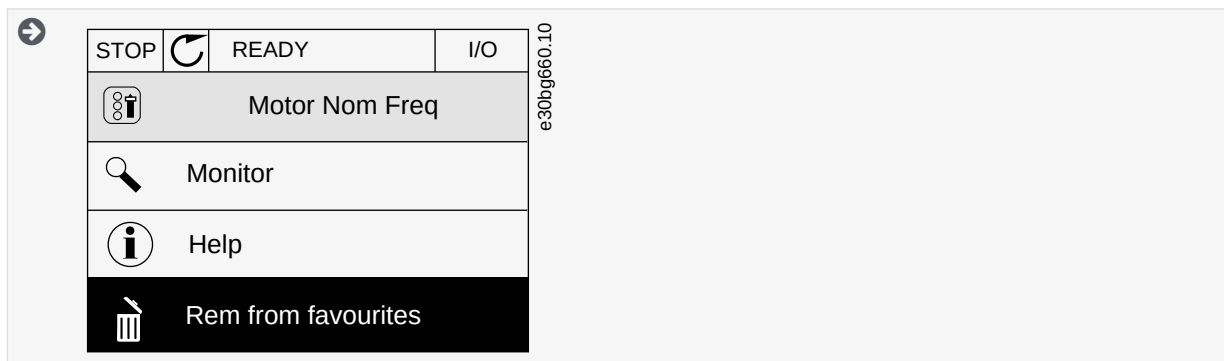
3. To continue, read the instructions on the display.

9.2.3 Removing an Item from the Favourites

Procedure

1. Go to the *Favourites*.
2. Find the item to be removed. Push the [OK] button.

3. Select *Rem* from favourites.



4. To remove the item, push the [OK] button again.

9.3 User Levels

9.3.1 Using the User Levels Menu

Use the *User level* parameters to keep the personnel who are not approved from changing the parameters. Also accidental changes in the parameters can be prevented.

When a user level is selected, the user cannot see all the parameters on the display of the control panel.

9.3.2 The User Level Parameters

Table 127: The User Level Parameters

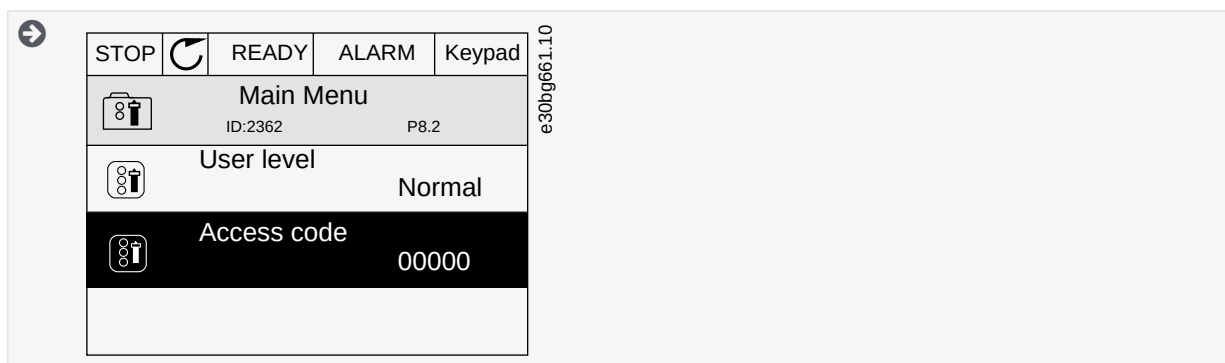
In- dex	Param- eter	Min	Max	Unit	De- fault	ID	Description
P8.1	User Level	1	3		1	1194	1 = Normal. All the menus are visible in the main menu. 2 = Monitoring. Only the monitoring and user level menus are visible in the main menu. 3 = Favourites. Only the favourites and user level menus are visible in the main menu. 4 = Monitoring & Favourites. Monitoring, favourites and user level menus are visible in the main menu.
P8.2	Access Code	0	99999		0	2362	If the value is set to be other than 0 before going to Monitoring from, for example, Normal, the access code has to be given when going back to Normal. This prevents personnel who are not approved from making changes in the parameters on the control panel. NOTE! Do not lose the access code. If the access code is lost, contact your nearest service center or partner.

9.3.3 Changing the Access Code of the User Levels

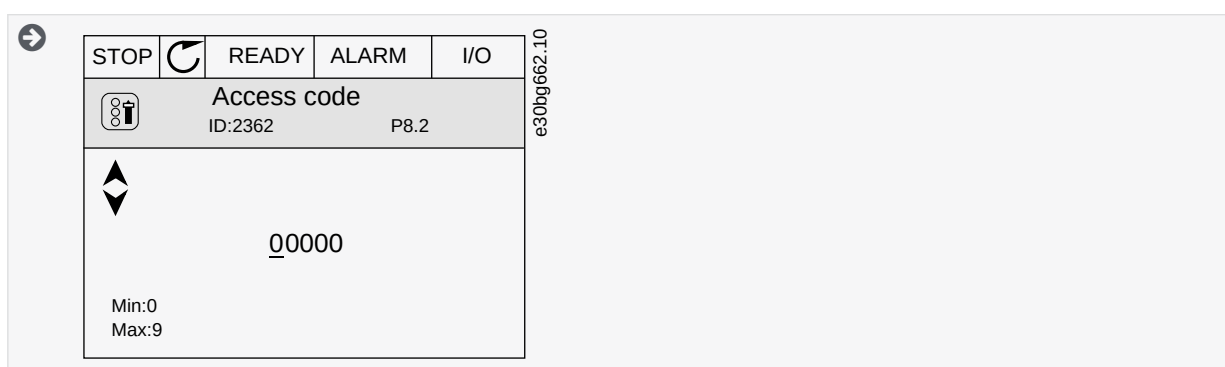
Procedure

1. Go to the *User levels*.

- Go to the item *Access code* and push the arrow button Right.



- To change the digits of the *Access code*, use all the arrow buttons.



- Accept the change with the [OK] button.

10 Monitoring Value Descriptions

10.1 Multimonitor

10.1.1 (ID 25) Frequency Reference

- Location in the menu with default settings: **V2.1.1** (In *Multimonitor* menu)
- Location in the menu: **V2.3.2** (In *Basic* menu)

This monitoring value shows the actual frequency reference to the motor control.
The value is updated at 10 ms interval.

10.1.2 (ID 1) Output Frequency

- Location in the menu with default settings: **V2.1.2** (In *Multimonitor* menu)
- Location in the menu: **V2.3.1** (In *Basic* menu)

This monitoring value shows the actual output frequency to the motor.

10.1.3 (ID 3) Motor Current

- Location in the menu with default settings: **V2.1.3** (In *Multimonitor* menu)
- Location in the menu: **V2.3.4** (In *Basic* menu)

This monitoring value shows the measured current of the motor.
The scaling of the value is different for different drive sizes.

10.1.4 (ID 2) Motor Speed

- Location in the menu with default settings: **V2.1.4** (In *Multimonitor* menu)
- Location in the menu: **V2.3.3** (In *Basic* menu)

This monitoring value shows the actual speed of the motor in RPM (calculated value).

10.1.5 (ID 5) Motor Shaft Power

- Location in the menu with default settings: **V2.1.5** (In *Multimonitor* menu)
- Location in the menu: **V2.3.7** (In *Basic* menu)

This monitoring value shows the actual shaft power of the motor (calculated value) as a percentage of the motor nominal power.

10.1.6 (ID 8) Unit Temperature

- Location in the menu with default settings: **V2.1.6** (In *Multimonitor* menu)
- Location in the menu: **V2.3.11** (In *Basic* menu)

This monitoring value shows the measured heat sink temperature of the drive.
The unit of measurement is degrees Celsius or Fahrenheit, depending on parameter C/F Selection (ID 1197) value.

10.1.7 (ID 7) DC Link Voltage

- Location in the menu with default settings: **V2.1.7** (In *Multimonitor* menu)
- Location in the menu: **V2.3.10** (In *Basic* menu)

This monitoring value shows the measured voltage in the DC link of the drive.

10.1.8 (ID 4) Motor Torque

- Location in the menu with default settings: **V2.1.8** (In *Multimonitor* menu)
- Location in the menu: **V2.3.5** (In *Basic* menu)

This monitoring value shows the actual torque of the motor (calculated value).

10.1.9 (ID 6) Motor Voltage

- Location in the menu with default settings: **V2.1.9** (In *Multimonitor* menu)
- Location in the menu: **V2.3.9** (In *Basic* menu)

This monitoring value shows the actual output voltage to the motor.

10.2 Basic

10.2.1 (ID 1) Output Frequency

- Location in the menu with default settings: **V2.1.2** (In *Multimonitor* menu)
- Location in the menu: **V2.3.1** (In *Basic* menu)

This monitoring value shows the actual output frequency to the motor.

10.2.2 (ID 25) Frequency Reference

- Location in the menu with default settings: **V2.1.1** (In *Multimonitor* menu)
- Location in the menu: **V2.3.2** (In *Basic* menu)

This monitoring value shows the actual frequency reference to the motor control.

The value is updated at 10 ms interval.

10.2.3 (ID 2) Motor Speed

- Location in the menu with default settings: **V2.1.4** (In *Multimonitor* menu)
- Location in the menu: **V2.3.3** (In *Basic* menu)

This monitoring value shows the actual speed of the motor in RPM (calculated value).

10.2.4 (ID 3) Motor Current

- Location in the menu with default settings: **V2.1.3** (In *Multimonitor* menu)
- Location in the menu: **V2.3.4** (In *Basic* menu)

This monitoring value shows the measured current of the motor.

The scaling of the value is different for different drive sizes.

10.2.5 (ID 4) Motor Torque

- Location in the menu with default settings: **V2.1.8** (In *Multimonitor* menu)
- Location in the menu: **V2.3.5** (In *Basic* menu)

This monitoring value shows the actual torque of the motor (calculated value).

10.2.6 (ID 5) Motor Shaft Power

- Location in the menu with default settings: **V2.1.5** (In *Multimonitor* menu)
- Location in the menu: **V2.3.7** (In *Basic* menu)

This monitoring value shows the actual shaft power of the motor (calculated value) as a percentage of the motor nominal power.

10.2.7 (ID 73) Motor Shaft Power

Location in the menu: **V2.3.8**

This monitoring value shows the actual shaft power of the motor (calculated value). The unit of measurement is kW or hp, depending on parameter kW/hp Selection (ID 1198) value.

The number of decimals in the value of this monitoring value varies depending on the size of the AC drive. In fieldbus control, ID 15592 can be mapped as Process Data Out to determine how many decimals are used. The last significant digit tells the amount of decimals.

10.2.8 (ID 6) Motor Voltage

- Location in the menu with default settings: **V2.1.9** (In *Multimonitor* menu)
- Location in the menu: **V2.3.9** (In *Basic* menu)

This monitoring value shows the actual output voltage to the motor.

10.2.9 (ID 7) DC Link Voltage

- Location in the menu with default settings: **V2.1.7** (In *Multimonitor* menu)
- Location in the menu: **V2.3.10** (In *Basic* menu)

This monitoring value shows the measured voltage in the DC link of the drive.

10.2.10 (ID 8) Unit Temperature

- Location in the menu with default settings: **V2.1.6** (In *Multimonitor* menu)
- Location in the menu: **V2.3.11** (In *Basic* menu)

This monitoring value shows the measured heat sink temperature of the drive.

The unit of measurement is degrees Celsius or Fahrenheit, depending on parameter C/F Selection (ID 1197) value.

10.2.11 (ID 9) Motor Temperature

Location in the menu: **V2.3.12**

This monitoring value shows the calculated motor temperature in percentage of the nominal working temperature.

When the value rises above 105%, motor thermal protection fault occurs.

10.2.12 (ID 1228) Motor Preheat

Location in the menu: **V2.3.13**

This monitoring value shows the status of the motor preheat function.

10.2.13 (ID 1054) kWh Trip Counter Low

Location in the menu: **V2.3.15**

This monitoring value shows the actual value of the kWh counter (energy counter). When the value of the counter goes higher than 65535, the counter is restarted from 0.

10.2.14 (ID 1067) kWh Trip Counter High

Location in the menu: **V2.3.16**

This monitoring value shows how many times the kWh counter (energy counter) has spun around.

10.3 I/O

10.3.1 (ID 15) SlotA DIN 1,2,3

Location in the menu: **V2.4.1**

This monitoring value shows the status of the digital inputs 1–3 in slot A (standard I/O).

10.3.2 (ID 16) SlotA DIN 4,5,6

Location in the menu: **V2.4.2**

This monitoring value shows the status of the digital inputs 4–6 in slot A (standard I/O).

10.3.3 (ID 17) SlotB RO 1,2,3

Location in the menu: **V2.4.3**

This monitoring value shows the status of the relay outputs 1–3 in slot B.

10.3.4 (ID 59) Analog Input 1

Location in the menu: **V2.4.4**

This monitoring value shows the value of the analog input signal as a percentage of the used range.

10.3.5 (ID 60) Analog Input 2

Location in the menu: **V2.4.5**

This monitoring value shows the value of the analog input signal as a percentage of the used range.

10.3.6 (ID 61) Analog Input 3

Location in the menu: **V2.4.6**

This monitoring value shows the value of the analog input signal as a percentage of the used range.

10.3.7 (ID 62) Analog Input 4

Location in the menu: **V2.4.7**

This monitoring value shows the value of the analog input signal as a percentage of the used range.

10.3.8 (ID 75) Analog Input 5

Location in the menu: **V2.4.8**

This monitoring value shows the value of the analog input signal as a percentage of the used range.

10.3.9 (ID 76) Analog Input 6

Location in the menu: **V2.4.9**

This monitoring value shows the value of the analog input signal as a percentage of the used range.

10.3.10 (ID 81) SlotA AO 1

Location in the menu: **V2.4.10**

This monitoring value shows the value of the analog output as a percentage of the used range.

10.4 Temperature Inputs

The monitoring values related to temperature input settings are only available if a B8 or BH option board is installed.

10.4.1 (ID 50) Temperature Input 1

Location in the menu: **V2.5.1i**

This monitoring value shows the measured value of the temperature. The unit of the monitoring value is degrees Celsius or Fahrenheit, depending on parameter C/F Selection (ID 1197) value.

The list of temperature inputs is made of the first 6 available temperature inputs. The list starts from slot A and ends in slot E. If an input is available but no sensor is connected, the list shows the maximum value because the measured resistance is endless. To make the value go to its minimum value, hardwire the input.

10.4.2 (ID 51) Temperature Input 2

Location in the menu: **V2.5.2**

This monitoring value shows the measured value of the temperature. The unit of the monitoring value is degrees Celsius or Fahrenheit, depending on parameter C/F Selection (ID 1197) value.

10.4.3 (ID 52) Temperature Input 3

Location in the menu: **V2.5.3**

This monitoring value shows the measured value of the temperature. The unit of the monitoring value is degrees Celsius or Fahrenheit, depending on parameter C/F Selection (ID 1197) value.

10.4.4 (ID 69) Temperature Input 4

Location in the menu: **V2.5.4**

This monitoring value shows the measured value of the temperature. The unit of the monitoring value is degrees Celsius or Fahrenheit, depending on parameter C/F Selection (ID 1197) value.

10.4.5 (ID 70) Temperature Input 5

Location in the menu: **V2.5.5**

This monitoring value shows the measured value of the temperature. The unit of the monitoring value is degrees Celsius or Fahrenheit, depending on parameter C/F Selection (ID 1197) value.

10.4.6 (ID 71) Temperature Input 6

Location in the menu: **V2.5.6**

This monitoring value shows the measured value of the temperature. The unit of the monitoring value is degrees Celsius or Fahrenheit, depending on parameter C/F Selection (ID 1197) value.

10.5 Extras and Advanced

10.5.1 (ID 43) Drive Status Word

Location in the menu: **V2.6.1**

This monitoring value shows the bit-coded status of the drive.

10.5.2 (ID 78) Ready Status

Location in the menu: **V2.6.2**

This monitoring value shows the bit-coded data about the Ready criteria of the drive. This data is useful for monitoring when the drive is not in the Ready state.

NOTICE

The values are visible as checkboxes on the graphical display. If a box is selected, the value is active.

10.5.3 (ID 89) Application Status Word 1

Location in the menu: **V2.6.3**

This monitoring value shows the bit-coded statuses of the application.

NOTICE

The values are visible as checkboxes on the graphical display. If a box is selected, the value is active.

10.5.4 (ID 90) Application Status Word 2

Location in the menu: **V2.6.4**

This monitoring value shows the bit-coded statuses of the application.

NOTICE

The values are visible as checkboxes on the graphical display. If a box is selected, the value is active.

10.5.5 (ID 56) DIN Status Word 1

Location in the menu: **V2.6.5**

This monitoring value shows the bit-coded status of the digital input signals.

The monitoring value is a 16-bit word, where each bit shows the status of 1 digital input. From each slot, 6 digital inputs are read. Word 1 starts from the input 1 in slot A (bit0) and ends with input 4 in slot C (bit15).

10.5.6 (ID 57) DIN Status Word 2

Location in the menu: **V2.6.6**

This monitoring value shows the bit-coded status of the digital input signals.

The monitoring value is a 16-bit word, where each bit shows the status of 1 digital input. From each slot, 6 digital inputs are read. Word 2 starts from the input 5 in slot C (bit0) and ends with input 6 in slot E (bit13).

10.5.7 (ID 45) Motor Current 1 Deci

Location in the menu: **V2.6.7**

This monitoring value shows the measured current of the motor with the fixed number of decimals and that is less filtered. This monitoring value can be used, for example, with fieldbus to get the correct value so that the enclosure size does not have an effect. It can also be used for monitoring when less filtering time is needed for the motor current.

10.5.8 (ID 1495) Frequency Reference Source

Location in the menu: **V2.6.8**

This monitoring value shows the momentary frequency reference source.

10.5.9 (ID 37) Last Active Fault Code

Location in the menu: **V2.6.9**

This monitoring value shows the fault code of latest activated fault that is not reset.

10.5.10 (ID 95) Last Active Fault ID

Location in the menu: **V2.6.10**

This monitoring value shows the fault ID of latest activated fault that is not reset.

10.5.11 (ID 74) Last Active Alarm Code

Location in the menu: **V2.6.11**

This monitoring value shows the alarm code of latest activated alarm that is not reset.

10.5.12 (ID 94) Last Active Alarm ID

Location in the menu: **V2.6.12**

This monitoring value shows the alarm ID of latest activated alarm that is not reset.

10.5.13 (ID 77) Motor Regulator Status

Location in the menu: **V2.6.13**

This monitoring value shows the bit-coded status of the motor limit controllers.

NOTICE

The values are visible as checkboxes on the graphical display. If a box is selected, the limit controller is active.

10.5.14 (ID 98) Motor Shaft Power 1 Decimal

Location in the menu: **V2.6.14**

This monitoring value shows the actual shaft power of the motor (calculated value with one decimal). The unit of measurement is kW or hp, depending on parameter kW/hp Selection (ID 1198) value.

10.6 Timer Functions

10.6.1 (ID 1441) TC 1, TC 2, TC 3

Location in the menu: **V2.7.1**

This monitoring value shows the status of the time channels 1, 2 and 3.

10.6.2 (ID 1442) Interval 1

Location in the menu: **V2.7.2**

This monitoring value shows the status of the interval function.

10.6.3 (ID 1443) Interval 2

Location in the menu: **V2.7.3**

This monitoring value shows the status of the interval function.

10.6.4 (ID 1444) Interval 3

Location in the menu: **V2.7.4**

This monitoring value shows the status of the interval function.

10.6.5 (ID 1445) Interval 4

Location in the menu: **V2.7.5**

This monitoring value shows the status of the interval function.

10.6.6 (ID 1446) Interval 5

Location in the menu: **V2.7.6**

This monitoring value shows the status of the interval function.

10.6.7 (ID 1447) Timer 1

Location in the menu: **V2.7.7**

The monitoring value shows the remaining time on the timer if the timer is active.

10.6.8 (ID 1448) Timer 2

Location in the menu: **V2.7.8**

The monitoring value shows the remaining time on the timer if the timer is active.

10.6.9 (ID 1449) Timer 3

Location in the menu: **V2.7.9**

The monitoring value shows the remaining time on the timer if the timer is active.

10.6.10 (ID 1450) Real-time Clock

Location in the menu: **V2.7.10**

This monitoring value shows the actual time of the real-time clock in a format of hh:mm:ss.

10.7 PID Controller

10.7.1 (ID 20) PID Setpoint

Location in the menu: **V2.8.1**

This monitoring value shows the value of the PID setpoint signal in process units. Use the parameter P3.13.1.7 to select the process unit (See [11.15.1.4 \(ID 1036\) Process Unit Selection](#)).

10.7.2 (ID 21) PID Feedback

Location in the menu: **V2.8.2**

This monitoring value shows the value of the PID feedback signal in process units. Use the parameter P3.13.1.7 to select the process unit (See [11.15.1.4 \(ID 1036\) Process Unit Selection](#)).

10.7.3 (ID 15541) PID Feedback (1)

Location in the menu: **V2.8.3**

This monitoring value shows the value of the PID feedback signal 1 in process units.

10.7.4 (ID 15542) PID Feedback (2)

Location in the menu: **V2.8.4**

This monitoring value shows the value of the PID feedback signal 2 in process units.

10.7.5 (ID 22) PID Error

Location in the menu: **V2.8.5**

This monitoring value shows the error value of the PID controller.

The error value is the deviation of PID feedback from the PID setpoint in process unit. Use the parameter P3.13.1.7 to select the process unit (See [11.15.1.4 \(ID 1036\) Process Unit Selection](#)).

10.7.6 (ID 23) PID Output

Location in the menu: **V2.8.6**

This monitoring value shows the output of the PID controller as a percentage (0–100%).

Give this value to the motor control (frequency reference) or to an analog output.

10.7.7 (ID 24) PID Status

Location in the menu: **V2.8.7**

This monitoring value shows the state of the PID controller.

10.8 External PID Controller

10.8.1 (ID 83) EXTPID Setpoint

Location in the menu: **V2.9.1**

This monitoring value shows the value of the PID setpoint signal in process units. Use the parameter P3.14.1.10 to select the process unit (See [11.15.1.7 \(ID 1035\) Process Unit Decimals](#)).

10.8.2 (ID 85) EXTPID Error

Location in the menu: **V2.9.3**

This monitoring value shows the error value of the PID controller. The error value is the deviation of PID feedback from the PID setpoint in process unit. Use the parameter P3.14.1.10 to select the process unit (See [11.15.1.7 \(ID 1035\) Process Unit Decimals](#)).

10.8.3 (ID 84) EXTPID Feedback

Location in the menu: **V2.9.2**

This monitoring value shows the value of the PID feedback signal in process units. Use the parameter P3.14.1.10 to select the process unit (See [11.15.1.7 \(ID 1035\) Process Unit Decimals](#)).

10.8.4 (ID 86) EXTPID Output

Location in the menu: **V2.9.4**

This monitoring value shows the output of the PID controller as a percentage (0–100%). Give this value to, for example, the analog output.

10.8.5 (ID 87) EXTPID Status

Location in the menu: **V2.9.5**

This monitoring value shows the state of the PID controller.

10.9 Multi-pump

10.9.1 (ID 30) Motors Running

Location in the menu: **V2.10.1**

This monitoring value shows the actual number of motors that operate in the Multi-pump system.

10.9.2 (ID 1114) Autochange

Location in the menu: **V2.10.2**

This monitoring value shows the status of the autochange requested.

10.9.3 (ID 1503) Next Autochange

Location in the menu: **V2.10.3**

This monitoring value shows the time that remains to next autochange.

10.9.4 (ID 1505) Operate Mode

Location in the menu: **V2.10.4**

This monitoring value shows the operation mode of the drive in the Multi-pump system.

10.9.5 (ID 1628) Multi-pump Status

Location in the menu: **V2.10.5**

This monitoring value shows the status of the drive in the Multi-pump system.

10.9.6 (ID 1629) Communication Status

Location in the menu: **V2.10.6**

This monitoring value shows status of the communication between the drives in the Multi-pump system.

10.9.7 (ID 1620) Pump (1) Running Time

Location in the menu: **V2.10.7**

This monitoring value shows the operating hours of the pump in Multi-pump system.

10.9.8 (ID 1621) Pump (2) Running Time

Location in the menu: **V2.10.8**

This monitoring value shows the operating hours of the pump in Multi-pump system.

10.9.9 (ID 1622) Pump (3) Running Time

Location in the menu: **V2.10.9**

This monitoring value shows the operating hours of the pump in Multi-pump system.

10.9.10 (ID 1623) Pump (4) Running Time

Location in the menu: **V2.10.10**

This monitoring value shows the operating hours of the pump in Multi-pump system.

10.9.11 (ID 1624) Pump (5) Running Time

Location in the menu: **V2.10.11**

This monitoring value shows the operating hours of the pump in Multi-pump system.

10.9.12 (ID 1625) Pump (6) Running Time

Location in the menu: **V2.10.12**

This monitoring value shows the operating hours of the pump in Multi-pump system.

10.9.13 (ID 1626) Pump (7) Running Time

Location in the menu: **V2.10.13**

This monitoring value shows the operating hours of the pump in Multi-pump system.

10.9.14 (ID 1627) Pump (8) Running Time

Location in the menu: **V2.10.14**

This monitoring value shows the operating hours of the pump in Multi-pump system.

10.10 Maintenance Counters

10.10.1 (ID 1101) Maintenance Counter 1

Location in the menu: **V2.11.1**

This monitoring value shows the status of the maintenance counter. The status of the maintenance counter is shown as revolutions multiplied by 1000, or in hours. For the configuration and activation of this counter, see [11.18.1 The Maintenance Counters](#).

10.11 Fieldbus Data

10.11.1 (ID 874) FB Control Word

Location in the menu: **V2.12.1**

This monitoring value shows the status of the fieldbus control word that the application uses in bypass mode. Depending on the fieldbus type or profile, the data that is received from the fieldbus can be modified before it is sent to the application.

Bit	Value = 0 (FALSE)	Value = 1 (TRUE)
Bit 0	Stop request from Fieldbus	Start request from Fieldbus
Bit 1	Forward direction request	Reverse direction request
Bit 2	No action	Reset active faults and alarms (on rising edge 0->1)
Bit 3	No action	Force stop mode to Coasting
Bit 4	No action	Force stop mode to Ramping
Bit 5	No action (normal deceleration ramp time)	Force drive to use fast deceleration ramp time (1/3 of normal deceleration time)
Bit 6	No action	Freeze drive frequency reference
Bit 7	No action	Force Fieldbus frequency reference to zero
Bit 8	No action	Force drive control place to Fieldbus control
Bit 9	No action	Force drive reference source to Fieldbus reference
Bit 10	Reserved	Reserved
Bit 11	Reserved	Reserved
Bit 12	No action	Activate Quick Stop function NOTE! This stops the drive according to setting in parameter menu M3.9.5.
Bit 13	Reserved	Reserved
Bit 14	Reserved	Reserved
Bit 15	Reserved	Reserved

10.11.2 (ID 875) FB Speed Reference

Location in the menu: **V2.12.2**

This monitoring value shows the fieldbus frequency reference as a percentage of minimum frequency to maximum frequency.

The speed reference information is scaled between the minimum and the maximum frequency at the moment when the application received it. The minimum and the maximum frequencies can be changed after the application received the reference without an effect on the reference.

10.11.3 (ID 876) FB Data IN 1

Location in the menu: **V2.12.3**

This monitoring value shows the raw value of process data in a 32-bit signed format.

10.11.4 (ID 877) FB Data IN 2

Location in the menu: **V2.12.4**

This monitoring value shows the raw value of process data in a 32-bit signed format.

10.11.5 (ID 878) FB Data IN 3

Location in the menu: **V2.12.5**

This monitoring value shows the raw value of process data in a 32-bit signed format.

10.11.6 (ID 879) FB Data IN 4

Location in the menu: **V2.12.6**

This monitoring value shows the raw value of process data in a 32-bit signed format.

10.11.7 (ID 880) FB Data IN 5

Location in the menu: **V2.12.7**

This monitoring value shows the raw value of process data in a 32-bit signed format.

10.11.8 (ID 881) FB Data IN 6

Location in the menu: **V2.12.8**

This monitoring value shows the raw value of process data in a 32-bit signed format.

10.11.9 (ID 882) FB Data IN 7

Location in the menu: **V2.12.9**

This monitoring value shows the raw value of process data in a 32-bit signed format.

10.11.10 (ID 883) FB Data IN 8

Location in the menu: **V2.12.10**

This monitoring value shows the raw value of process data in a 32-bit signed format.

10.11.11 (ID 864) FB Status Word

Location in the menu: **V2.12.11**

This monitoring value shows the status of the fieldbus status word that the application uses in bypass mode. Depending on the fieldbus type or profile, the data can be modified before it is sent to the fieldbus.

Bit	Value = 0 (FALSE)	Value = 1 (TRUE)
Bit 0	Not ready to operate	Ready to operate
Bit 1	Not running	Running
Bit 2	Behavior depends on P3.6.17.	Behavior depends on P3.6.17.
Bit 3	No fault	Fault is active
Bit 4	No alarm	Alarm is active
Bit 5	Requested speed not reached	Running at requested speed
Bit 6	Actual speed of the drive not zero	Actual speed of the drive is zero
Bit 7	Motor not magnetized (flux not ready)	Motor magnetized (flux ready)
Bit 8	Reserved	Reserved
Bit 9	Reserved	Reserved
Bit 10	Reserved	Reserved
Bit 11	Reserved	Reserved
Bit 12	Reserved	Reserved
Bit 13	Reserved	Reserved
Bit 14	Reserved	Reserved
Bit 15	Reserved	Reserved

10.11.12 (ID 865) FB Speed Actual

Location in the menu: **V2.12.12**

This monitoring value shows the actual speed of the drive as a percentage of minimum frequency and maximum frequency. The value 0% indicates the minimum frequency and the value 100% indicates the maximum frequency. This monitoring value is continuously updated depending on the momentary min and max frequencies and the output frequency.

10.11.13 (ID 866) FB Data OUT 1Location in the menu: **V2.12.13**

This monitoring value shows the raw value of process data in a 32-bit signed format.

10.11.14 (ID 867) FB Data OUT 2Location in the menu: **V2.12.13**

This monitoring value shows the raw value of process data in a 32-bit signed format.

10.11.15 (ID 868) FB Data OUT 3Location in the menu: **V2.12.15**

This monitoring value shows the raw value of process data in a 32-bit signed format.

10.11.16 (ID 869) FB Data OUT 4Location in the menu: **V2.12.16**

This monitoring value shows the raw value of process data in a 32-bit signed format.

10.11.17 (ID 870) FB Data OUT 5Location in the menu: **V2.12.17**

This monitoring value shows the raw value of process data in a 32-bit signed format.

10.11.18 (ID 871) FB Data OUT 6Location in the menu: **V2.12.18**

This monitoring value shows the raw value of process data in a 32-bit signed format.

10.11.19 (ID 872) FB Data OUT 7Location in the menu: **V2.12.19**

This monitoring value shows the raw value of process data in a 32-bit signed format.

10.11.20 (ID 873) FB Data OUT 8Location in the menu: **V2.12.20**

This monitoring value shows the raw value of process data in a 32-bit signed format.

10.12 Drive Customizer**10.12.1 (ID 15020) Block Out.1**Location in the menu: **V2.13.2**

This monitoring value shows the value of the function block output in the Drive customizer function.

10.12.2 (ID 15040) Block Out.2Location in the menu: **V2.13.3**

This monitoring value shows the value of the function block output in the Drive customizer function.

10.12.3 (ID 15060) Block Out.3Location in the menu: **V2.13.4**

This monitoring value shows the value of the function block output in the Drive customizer function.

10.12.4 (ID 15080) Block Out.4Location in the menu: **V2.13.5**

This monitoring value shows the value of the function block output in the Drive customizer function.

10.12.5 (ID 15100) Block Out.5Location in the menu: **V2.13.6**

This monitoring value shows the value of the function block output in the Drive customizer function.

10.12.6 (ID 15120) Block Out.6

Location in the menu: **V2.13.7**

This monitoring value shows the value of the function block output in the Drive customizer function.

10.12.7 (ID 15140) Block Out.7

Location in the menu: **V2.13.8**

This monitoring value shows the value of the function block output in the Drive customizer function.

10.12.8 (ID 15160) Block Out.8

Location in the menu: **V2.13.9**

This monitoring value shows the value of the function block output in the Drive customizer function.

10.12.9 (ID 15180) Block Out.9

Location in the menu: **V2.13.10**

This monitoring value shows the value of the function block output in the Drive customizer function.

10.12.10 (ID 15200) Block Out.10

Location in the menu: **V2.13.11**

This monitoring value shows the value of the function block output in the Drive customizer function.

10.13 Condition-based Monitoring

10.13.1 Baseline Data

10.13.1.1 (ID 3601) Baseline Data Element 1

Location in the menu: **V2.16.1.1**

This monitoring value shows the baseline data. The baseline data is selected using the baseline data selector.

10.13.1.2 (ID 3602) Baseline Data Element 2

Location in the menu: **V2.16.1.2**

This monitoring value shows the baseline data. The baseline data is selected using the baseline data selector.

10.13.1.3 (ID 3603) Baseline Data Element 3

Location in the menu: **V2.16.1.3**

This monitoring value shows the baseline data. The baseline data is selected using the baseline data selector.

10.13.1.4 (ID 3604) Baseline Data Element 4

Location in the menu: **V2.16.1.4**

This monitoring value shows the baseline data. The baseline data is selected using the baseline data selector.

10.13.1.5 (ID 3605) Baseline Data Element 5

Location in the menu: **V2.16.1.5**

This monitoring value shows the baseline data. The baseline data is selected using the baseline data selector.

10.13.1.6 (ID 3606) Baseline Data Element 6

Location in the menu: **V2.16.1.6**

This monitoring value shows the baseline data. The baseline data is selected using the baseline data selector.

10.13.1.7 (ID 3607) Baseline Data Element 7

Location in the menu: **V2.16.1.7**

This monitoring value shows the baseline data. The baseline data is selected using the baseline data selector.